

PAEDIATRIC PATIENT CASE

ASTar-driven treatment adjustment >40 hours earlier than standard-of-care

Name: Lucas

Age: 0



A CASE OF ESCALATION

CLINICAL OVERVIEW

Lucas lives in a region with a high prevalence of Antimicrobial Resistance. He was admitted to the NICU due to cardiopathy and was placed on invasive mechanical ventilation.

During his time in the NICU, he began to display symptoms associated with sepsis, and following clinical examination it was determined that he had a bloodstream infection associated with a central venous catheter.

A blood sample was sent to the microbiology lab for pathogen ID and Antimicrobial Susceptibility Testing (AST).

Antibiotic	Breakpoints		ASTar		SoC (BMD)	
	S≤	R>	MIC	SIR	MIC	SIR
Meropenem	2	8	≤0.06	S	≤0.12	S
Piperacillin-Tazobactam	8	8	16	R	16*	R*

*BMD was repeated due to a discrepant first MIC result reported within Susceptible range

Empiric therapy

A broad spectrum therapy of intravenous Piperacillin-Tazobactam at a dosage of 100 mg/kg every 8 hours was initiated while the blood sample underwent testing. The patient's condition did not improve during this time.

ASTar-driven therapy

Results from the lab identified the pathogen as *Enterobacter cloacae* complex. ASTar delivered an actionable AST report to the clinical team **6 hours and 16 minutes** after the blood sample was loaded and ASTar identified pathogen resistance to the empiric Piperacillin-Tazobactam, a result misidentified by BMD. Treatment was escalated to intravenous Meropenem at a dosage of 30 mg/kg every 8 hours.

Patient outcomes

Meropenem treatment was ended after 14.5 days following clinical cure of the infection. Lucas suffered no side effects from the antibiotics and was transferred to the cardiology ward for further care. He was discharged from the hospital three months later.

ASTar guided treatment adjustment **40.25 hours sooner than standard-of-care** (BMD) methods.

This patient case is a real case from clinical studies.
The name has been changed to protect patient identity.

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